

12. Players of “bubble soccer” wear air-filled plastic “bubbles” as shown.



Which of the following statements best explains why the bubble can reduce the chance of injury during a collision ?

- A. The bubble increases the mass of the player, thus the momentum of the player increases.
- B. The bubble increases the air resistance acting on the player.
- C. The bubble lengthens the impact time during a collision.
- D. Like a balloon, the bubble provides a lifting force to the player.

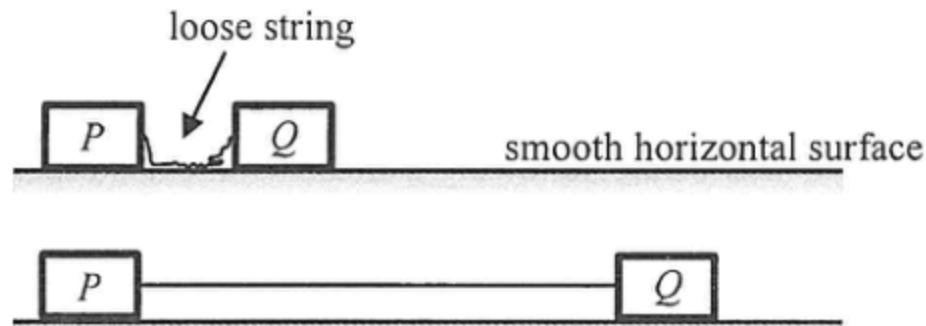
7. An egg is released from a height and falls towards a cushion without breaking.



Which of the following is the most probable explanation ?

- A. The cushion makes the egg stop at a shorter distance.
- B. The cushion helps lengthen the time of impact.
- C. The cushion helps reduce part of the gravitational force acting on the egg.
- D. The cushion increases the reaction force acting on the egg during impact.

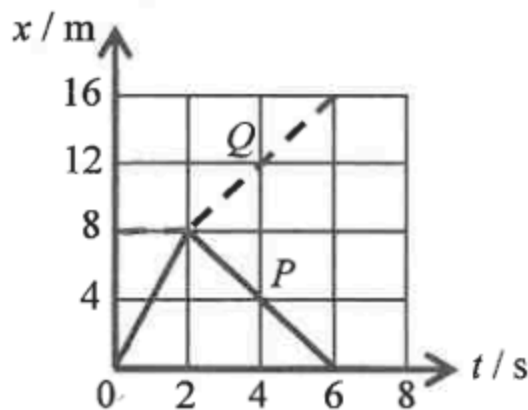
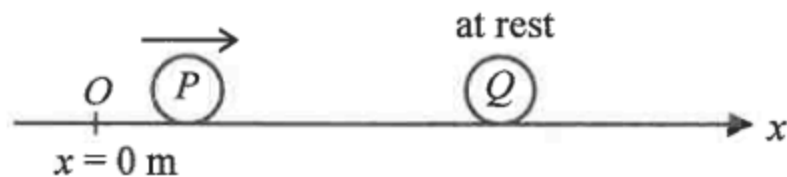
8. On a smooth horizontal surface, two identical blocks P and Q are connected by a light inextensible string. The blocks are at rest and the string is loose initially.



Q is given a speed of 4 m s^{-1} and moves to the right. Find the speeds of the blocks when the string just becomes taut and P starts to move.

	block P	block Q
A.	1 m s^{-1}	1 m s^{-1}
B.	2 m s^{-1}	1 m s^{-1}
C.	2 m s^{-1}	2 m s^{-1}
D.	4 m s^{-1}	2 m s^{-1}

12.



On a smooth horizontal surface, sphere P travelling along the x -axis collides head-on with another sphere Q , which is initially at rest at $x = 8 \text{ m}$. The graph shows the displacement-time ($x-t$) relationship of P (solid line) and Q (dotted line). The collision takes place at $t = 2 \text{ s}$ and the collision time is negligible. Which statement is correct?

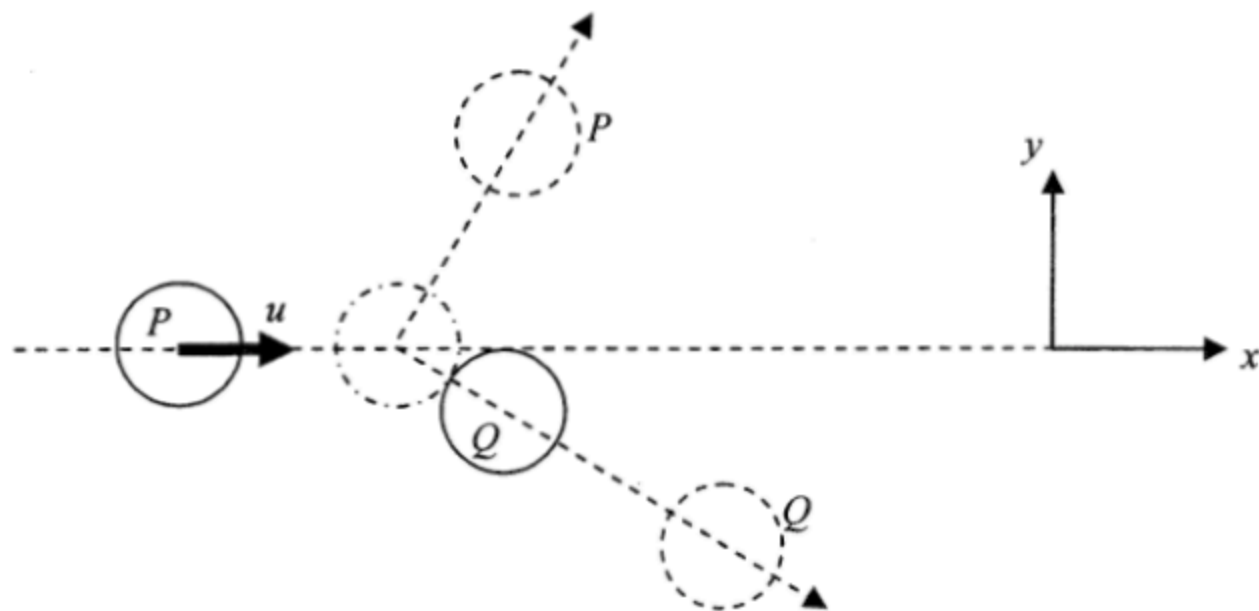
- A. The collision is perfectly inelastic.
- B. The mass of P is larger than that of Q .
- C. P still moves along $+x$ direction after collision.
- D. P and Q move with the same speed after collision.

12. At a certain instant, an object flying horizontally to the right at 1 m s^{-1} suddenly explodes into two fragments of mass ratio $1 : 2$. If just after the explosion the more massive fragment flies to the right at 3 m s^{-1} , what would happen to the other fragment just after the explosion ?

- A. It would fly to the left at 3 m s^{-1} .
- B. It would fly to the left at 4 m s^{-1} .
- C. It would be momentarily at rest.
- D. It would fly to the right at 1 m s^{-1} .

13.

On a smooth horizontal surface, a circular disk P moving at velocity u along the x direction collides obliquely with an identical disk Q initially at rest as shown below. The mass of each disk is m . Which statements about the collision is/are correct ?



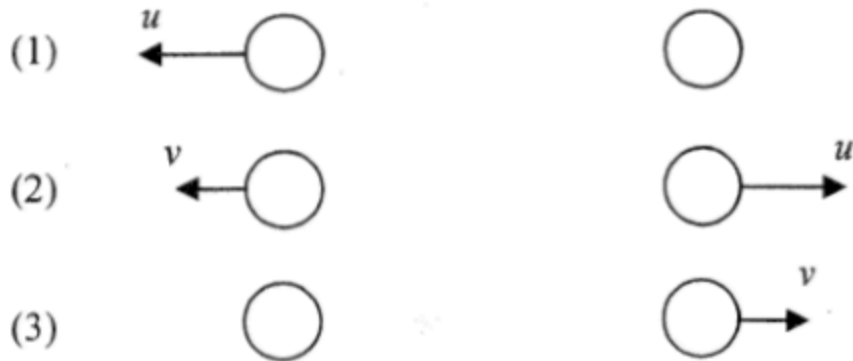
- (1) Momentum of the system in the y direction is not conserved.
- (2) The total kinetic energy of P and Q after collision is $\frac{1}{2}mu^2$ if the collision is perfectly elastic.
- (3) Speed of Q after collision is less than u .

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

10.

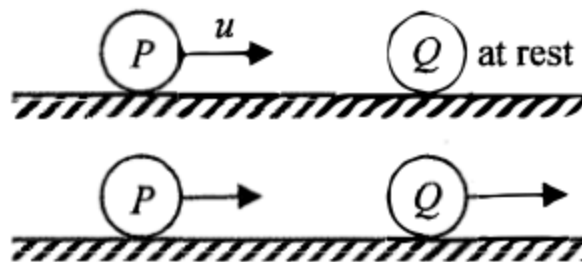


Two identical spheres are moving in opposite directions with speeds u and v (with $u > v$) respectively as shown. They make a head-on collision. Which of the following diagrams show(s) a *possible* situation of the spheres after collision?



- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

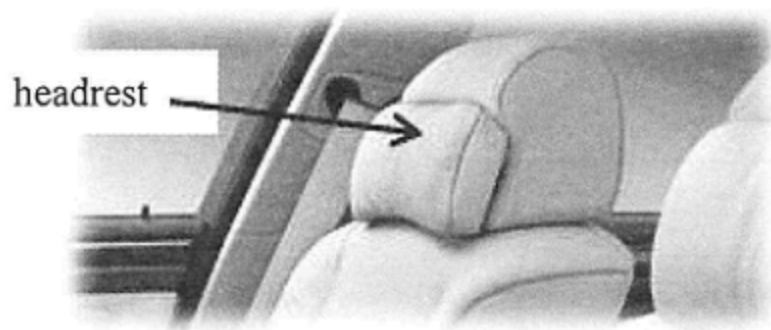
12. On a smooth horizontal surface, a marble P moving with speed u collides head-on with another marble Q , which is at rest. After collision, P and Q move with different speeds as shown.



Which of the following statements about this collision is/are correct ?

- (1) During collision, the force acting on Q by P is equal and opposite to that acting on P by Q .
 - (2) The total momentum of the two marbles is conserved only when the collision is perfectly elastic.
 - (3) The kinetic energy lost by P must be equal to that gained by Q .
- A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

5.

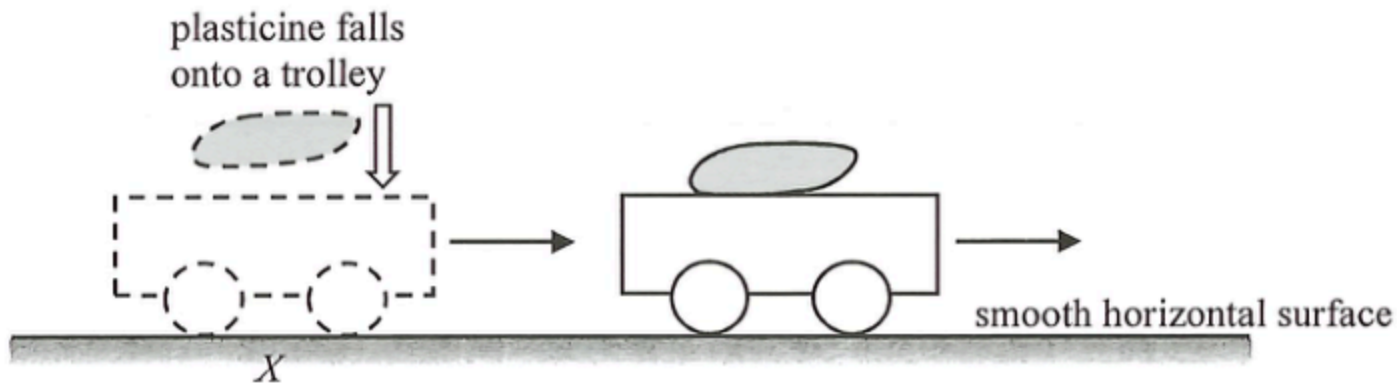


For a car travelling on a highway, which of the following statements about the safety design of the headrest is/are correct ?

- (1) As the headrest is soft, it can reduce the force exerted on the passenger's head during impact.
- (2) It can minimise injury of the passenger when the car is struck by another one from behind.
- (3) It can minimise injury of the passenger when the car brakes suddenly.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

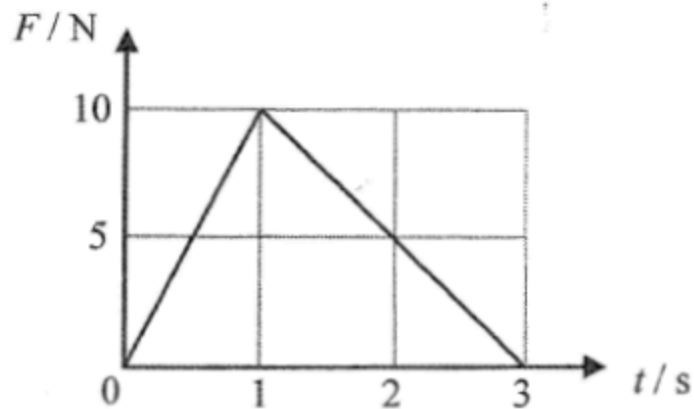
11. A trolley moves with a constant velocity along a smooth horizontal surface. When the trolley reaches point X , a plasticine falls vertically onto it. After the collision, they stick together and continue to move forward.



Which description about the **total linear momentum** of the trolley and the plasticine just before and just after collision is correct ?

- | | along the horizontal direction | along the vertical direction |
|----|--------------------------------|------------------------------|
| A. | conserved | conserved |
| B. | conserved | not conserved |
| C. | not conserved | conserved |
| D. | not conserved | not conserved |

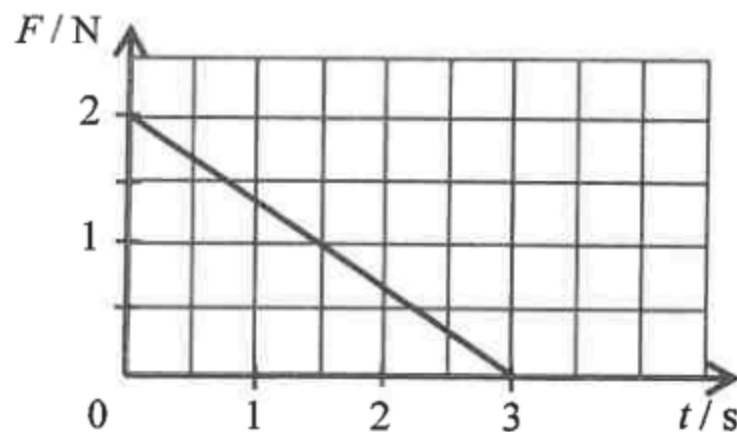
6.



An object of mass 3 kg is initially at rest on a smooth horizontal ground. A force F is applied horizontally to the object such that the magnitude of F varies with time t as shown. What is the speed of the object at $t = 3$ s? Neglect air resistance.

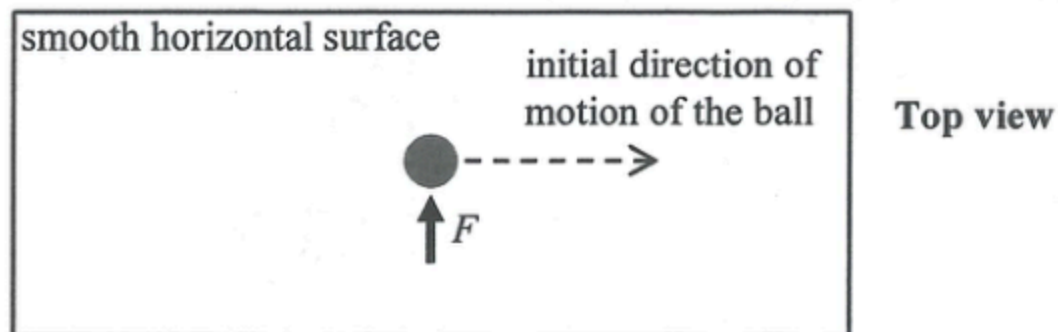
- A. 2.5 m s^{-1}
- B. 5 m s^{-1}
- C. 10 m s^{-1}
- D. 15 m s^{-1}

8. An object of mass 2 kg moves with velocity 1 m s^{-1} initially. It is acted upon by a force F which varies with time t as shown in the graph. The force and the object's velocity are in the same direction. Find the speed of the object at $t = 3\text{ s}$.



- A. 1.5 m s^{-1}
B. 2.5 m s^{-1}
C. 3.5 m s^{-1}
D. 4.5 m s^{-1}

10.



The above figure shows a ball moving with a constant speed along a straight line on a smooth horizontal surface. At a certain instant, the ball is acted on by a force F for a very short time as shown above. Which **subsequent** path below would the ball most closely follow ?

A.



B.



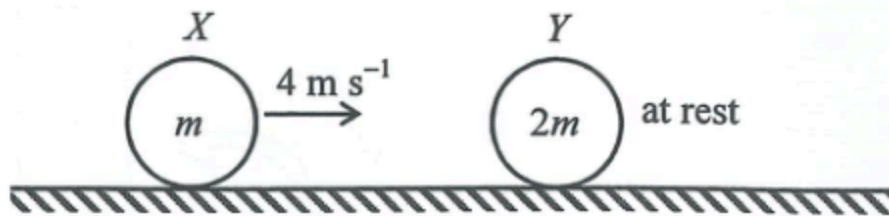
C.



D.



11.



On a smooth horizontal surface, sphere X of mass m travels with speed 4 m s^{-1} . It collides head-on with another sphere Y of mass $2m$, which is at rest initially. Which of the following can be the speed of Y just after collision?

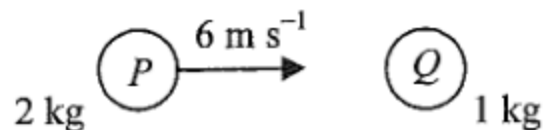
(1) 1 m s^{-1}

(2) 2 m s^{-1}

(3) 3 m s^{-1}

- A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

7.

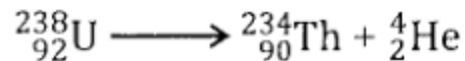


A sphere P of mass 2 kg makes a head-on collision with another sphere Q of mass 1 kg which is initially at rest. The speed of P just before collision is 6 m s^{-1} . If the two spheres move in the same direction after collision, which of the following could be the speed(s) of Q just after collision?

- (1) 2 m s^{-1}
- (2) 4 m s^{-1}
- (3) 6 m s^{-1}

- A. (1) only
- B. (1) and (2) only
- C. (2) and (3) only
- D. (1), (2) and (3)

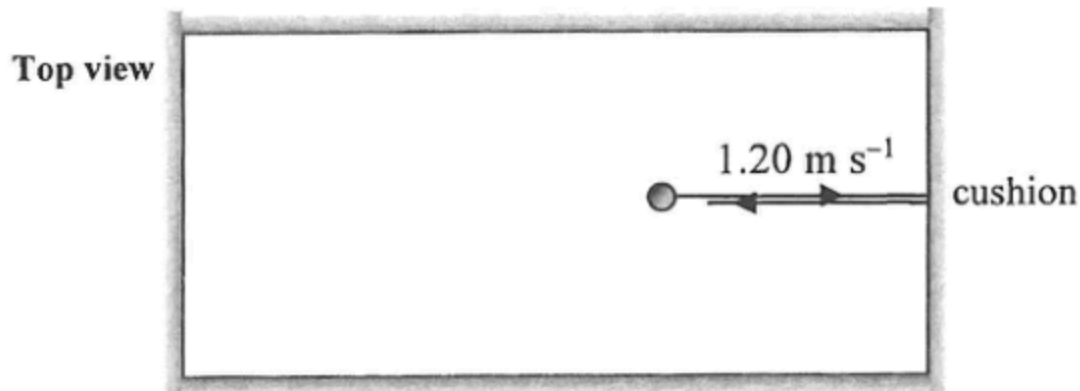
8. A stationary uranium nucleus ${}^{238}_{92}\text{U}$ decays to give a thorium nucleus ${}^{234}_{90}\text{Th}$ and an α particle ${}^4_2\text{He}$.



Which of the following correctly describes the situation about the ${}^{234}_{90}\text{Th}$ nucleus and the α particle just after the decay ?

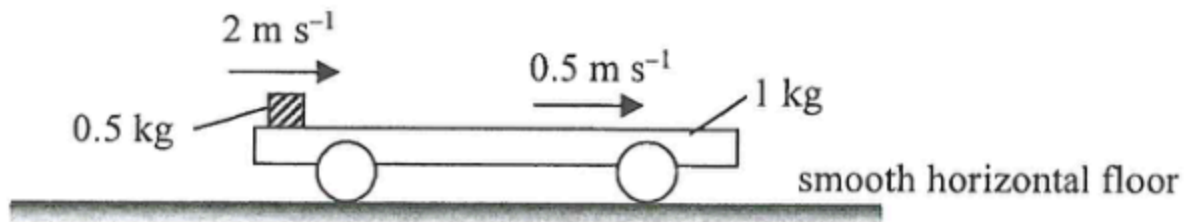
	magnitude of momentum p	kinetic energy KE
A.	$p(\text{Th}) = p(\alpha)$	$\text{KE}(\text{Th}) < \text{KE}(\alpha)$
B.	$p(\text{Th}) > p(\alpha)$	$\text{KE}(\text{Th}) > \text{KE}(\alpha)$
C.	$p(\text{Th}) = p(\alpha)$	$\text{KE}(\text{Th}) > \text{KE}(\alpha)$
D.	$p(\text{Th}) = p(\alpha)$	$\text{KE}(\text{Th}) = \text{KE}(\alpha)$

11. On a billiard table, a billiard ball of mass 0.16 kg and moving with a uniform speed of 1.20 m s^{-1} collides elastically with the cushion. It rebounds with the same speed normal to the cushion as shown. If the impact time is 0.0003 s , find the average force acting on the ball by the cushion during collision.



- A. 0 N
B. 640 N
C. 1280 N
D. 2560 N

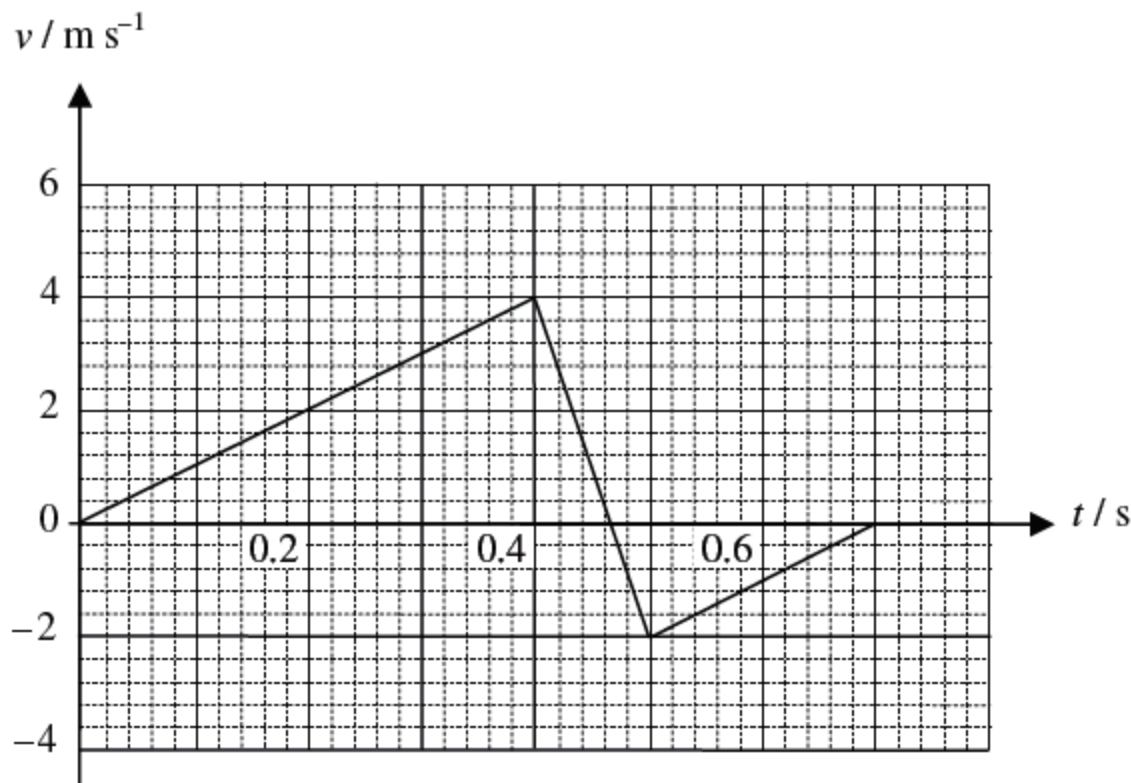
10. A 1-kg trolley with a 0.5-kg block placed on its horizontal platform is resting on a smooth horizontal floor. The block is then given a sharp push to the right. At an instant after the push, the block and the trolley are moving to the right at 2 m s^{-1} and 0.5 m s^{-1} respectively as observed from the floor.



Finally, they travel together with the same speed v . Find v . Neglect air resistance.

- A. 0.5 m s^{-1}
- B. 1 m s^{-1}
- C. 1.5 m s^{-1}
- D. 2 m s^{-1}

13.

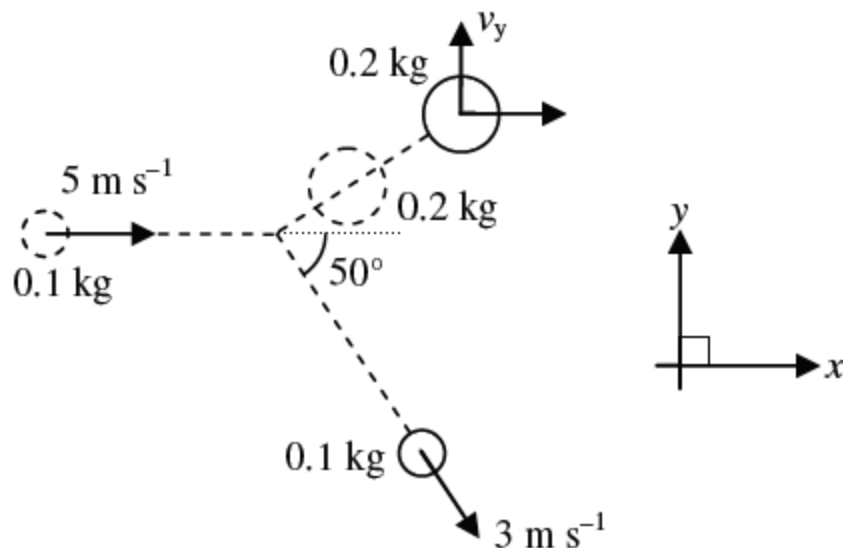


A ball of mass 0.2 kg is released from rest. It hits the ground and rebounds. The velocity-time graph of the ball is shown above. Which of the following statements are correct ?

- (1) The magnitude of the change in momentum of the ball during the collision is 1.2 kg m s^{-1} .
- (2) The magnitude of the average force acting on the ball by the ground during the collision is 12 N .
- (3) There is mechanical energy loss during the collision.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

- *14. A disc of mass 0.1 kg and velocity 5 m s^{-1} strikes a stationary disc of mass 0.2 kg on a smooth table. After the collision, the 0.1 kg disc moves with a speed of 3 m s^{-1} at 50° to the x direction. Find the component of the velocity of the 0.2 kg disc in y direction, v_y , after the collision.



- A. 1.15 m s^{-1}
- B. 1.54 m s^{-1}
- C. 1.92 m s^{-1}
- D. 2.01 m s^{-1}

13. A car P of mass 1000 kg moves with a speed of 20 m s^{-1} and makes a head-on collision with a car Q of mass 1500 kg, which was moving with a speed of 10 m s^{-1} in the opposite direction before the collision. The two cars stick together after the collision. Find their common velocity immediately after the collision.

- A. 2 m s^{-1} along the original direction of P
- B. 2 m s^{-1} along the original direction of Q
- C. 14 m s^{-1} along the original direction of P
- D. 14 m s^{-1} along the original direction of Q